DATA SHEET

PRODUCT **NTC Thermistor Sensor**

SERIES JAT Series

PART NO.

QUICK REFERENCE DATA

PARAMETER	VALUE	UNIT
Resistance Value R25	10~ 330	KΩ
B25/50	3380~4250	K
B25/85	3435~4360	K

ISSUE DATE 2023/2/4

REVISION DATE 2023/2/4

REFERENCE NO.

RoHS COMPLIANCE ITEM

Halogen Free

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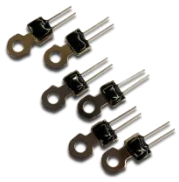
NTC Sensor Specialty JAT series



Features

RoHS / Halogen-Free (HF) compliant
Operating temperature range: -40℃~+125℃
Wide resistance range
Qualified based on AEC-Q200
Agency recognition: UL / TUV

符合 RoHS / Halogen-Free (HF)規範
工作溫度範圍：-40℃～+ 125℃
電阻範圍廣
符合 AEC-Q200
安規認證: UL / TUV

Applications

Home appliances
Office automation
Automotive
Switch mode power supplies
Adapters
Security

家電
OA 設備
汽車
開關模式電源
適配器
安防設備

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How to Order

Part Number Code																			
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
J	**A**	**T**	**1**	**0**	**3**	**F**	**3**	**4**	**4**	**F**	**B**	**5**	**2**	**0**	**0**	**1**	**0**	**3**	**X**
①			②			③	④			⑤	⑥	⑦	⑧	⑨					

①	Product Type	JAT series	⑤	Tolerance of B Value	F = ±1% G = ±2% H = ±3% J = ±5%	⑨	Optional Suffix	Internal Control Code
②	Zero Power Resistance @25℃（R25）	502 = 5KΩ 103 = 10KΩ 474 = 470KΩ	⑥	Definition of B Value	A = 25/50 B = 25/85			
③	Tolerance of R25	F = ±1% G = ±2% H = ±3% J = ±5% K = ±10%	⑦	Lead Diameter	5 = 0.5mm			
④	B Value	344 = 3435 K 405 = 4050 K	⑧	Lead spacing	2 = 2.5mm 4 = 3.5mm			

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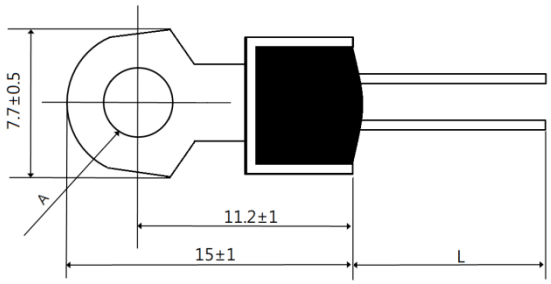
Structure and Dimension



L1: Designed by customer needs

Electrical Characteristics

Part No	Zero Power Resistance at 25℃	Tolerance of R25	B25/50 Value	Tolerance of B Value	Dissipation Factor	Thermal Time Constant	Max. Power Rating at 25℃	Safety Approvals	
	R 25 (Ω)	(± %)	(K)	(± %)	δ(mW/℃)	τ (sec.)	(mW)	cULus	
JAT103X338YA	10,000	10,5,3,2,1	3380	5,3,2,1	Approx. 2.5	Approx. 18	150	■	
JAT103X410YA	10,000	10,5,3,2,1	4100	5,3,2,1	Approx. 2.5	Approx. 18	150	■	
JAT473X395YA	47,000	10,5,3,2,1	3950	5,3,2,1	Approx. 2.5	Approx. 18	150	■	
JAT503X395YA	50,000	10,5,3,2,1	3950	5,3,2,1	Approx. 2.5	Approx. 18	150	■	
JAT104X395YA	100,000	10,5,3,2,1	3950	5,3,2,1	Approx. 2.5	Approx. 18	150		
JAT104X425YA	100,000	10,5,3,2,1	4250	5,3,2,1	Approx. 2.5	Approx. 18	150		

※ X : R Tolerance, Y : B Value Tolerance

Part No	Zero Power Resistance at 25℃	Tolerance of R25	B25/85 Value	Tolerance of B Value	Dissipation Factor	Thermal Time Constant	Max. Power Rating at 25 ℃	Safety Approvals	
	R 25 (Ω)	(± %)	(K)	(± %)	δ(mW/℃)	τ (sec.)	(mW)	cULus	
JAT103X344YB	10,000	10,5,3,2,1	3435	5,3,2,1	Approx. 2.5	Approx. 18	150	■	■
JAT103X398YB	10,000	10,5,3,2,1	3980	5,3,2,1	Approx. 2.5	Approx. 18	150	■	■
JAT333X398YB	33,000	10,5,3,2,1	3980	5,3,2,1	Approx. 2.5	Approx. 18	150		
JAT473X409YB	47,000	10,5,3,2,1	4090	5,3,2,1	Approx. 2.5	Approx. 18	150	■	
JAT503X409YB	50,000	10,5,3,2,1	4090	5,3,2,1	Approx. 2.5	Approx. 18	150	■	
JAT104X395YB	100,000	10,5,3,2,1	3950	5,3,2,1	Approx. 2.5	Approx. 18	150	■	
JAT104X408YB	100,000	10,5,3,2,1	4080	5,3,2,1	Approx. 2.5	Approx. 18	150	■	
JAT104X419YB	100,000	10,5,3,2,1	4190	5,3,2,1	Approx. 2.5	Approx. 18	150	■	
JAT104X425YB	100,000	10,5,3,2,1	4250	5,3,2,1	Approx. 2.5	Approx. 18	150		
JAT104X436YB	100,000	10,5,3,2,1	4360	5,3,2,1	Approx. 2.5	Approx. 18	150	■	
JAT204X406YB	200,000	10,5,3,2,1	4055	5,3,2,1	Approx. 2.5	Approx. 18	150	■	
JAT334X398YB	330,000	10,5,3,2,1	3980	5,3,2,1	Approx. 2.5	Approx. 18	150		

※ X : R Tolerance, Y : B Value Tolerance

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Reliability- NTC Sensor Specialty JAT series

<table>
<tr><th>Item</th><th>Standard</th><th>Test condition</th><th>Specifications</th></tr>
<tr><td>Terminal pull strength</td><td>IEC 60068-2-21</td><td>Gradually applying the force specified and keeping the unit fixed for 10±1 sec.
<table><tr><th>Terminal diameter (mm)</th><th>Force (Kg)</th></tr><tr><td>d ≦ 0.25mm</td><td>1N (0.102Kg)</td></tr><tr><td>0.25mm ＜ d ≦ 0.35mm</td><td>2.5N (0.255Kg)</td></tr><tr><td>0.35mm ＜ d ≦ 0.50mm</td><td>5N (0.510Kg)</td></tr><tr><td>0.50mm ＜ d ≦ 0.80mm</td><td>10N (1.02Kg)</td></tr></table></td><td>No visible damage</td></tr>
<tr><td>Bending Strength of Terminals</td><td>IEC 60068-2-21</td><td>Hold specimen and apply the force specified below to each lead.
Bend the specimen to 90° , then return to the original position.
Repeat the procedure in the opposite direction.
<table><tr><th>Terminal diameter (mm)</th><th>Force (Kg)</th></tr><tr><td>d ≦ 0.25mm</td><td>0.5N (0.051Kg)</td></tr><tr><td>0.25mm ＜ d ≦ 0.35mm</td><td>1.25N (0.128Kg)</td></tr><tr><td>0.35mm ＜ d ≦ 0.50mm</td><td>2.5N (0.255Kg)</td></tr><tr><td>0.50mm ＜ d ≦ 0.80mm</td><td>5N (0.510Kg)</td></tr></table></td><td>No visible damage</td></tr>
<tr><td>Solderability</td><td>IEC 60068-2-20</td><td>245±3℃, 3±0.3 sec</td><td>At least 95% of terminal electrode is covered by new solder</td></tr>
<tr><td>Resistance to soldering heat</td><td>IEC 60068-2-20</td><td>260±5℃,10±1 sec</td><td>No visible damage
△R25/R25≦±5%</td></tr>
<tr><td>High temperature storage</td><td>IEC 60068-2-2</td><td>125±2℃, 1000hrs</td><td>No visible damage
△R25/R25≦±5%</td></tr>
<tr><td>Damp Heat Steady State</td><td>IEC 60068-2-78</td><td>40±2℃ , 90~95% RH, 1000±24hrs</td><td>No visible damage
△R25/R25≦±5%</td></tr>
<tr><td>Rapid Change of Temperature</td><td>IEC 60068-2-14</td><td>The conditions shown below shall be repeated 5 cycles.
<table><tr><th>Step</th><th>Temperature (℃)</th><th>Period (minutes)</th></tr><tr><td>1</td><td>-40±5</td><td>30±3</td></tr><tr><td>2</td><td>Room temperature</td><td>5±3</td></tr><tr><td>3</td><td>125±5</td><td>30±3</td></tr><tr><td>4</td><td>Room temperature</td><td>5±3</td></tr></table></td><td>No visible damage
△R25/R25≦±5%</td></tr>
<tr><td>Life Test</td><td>IEC 60539-1 4.26.3</td><td>25±5℃ , Pmax, 1000hrs</td><td>No visible damage
△R25/R25≦±5%</td></tr>
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